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CPSC 332

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Study Spots Database: Project Design Report

1. Introduction

The goal of this project is to design and implement a relational database system that helps users find and review places suitable for studying. Rather than focusing exclusively on a single type of venue, the Study Spots database encompasses a wide range of locations, including cafes, libraries, university study rooms, and other places where people commonly go to study. These locations are collectively referred to as study spots throughout this report.

The motivation for selecting this topic stems from a common problem that students face: finding a good place to study can be surprisingly difficult. Different people have very different needs when it comes to study environments. Some people need absolute quiet, while others prefer a bit of background noise. Some need fast Wi-Fi and power outlets, while others just need a comfortable seat and good lighting. Currently, no single platform is specifically designed with the student studying experience in mind, and general review platforms like Yelp do not capture the study-relevant attributes that matter most to students.

The Study Spots database addresses this gap by organizing location data alongside study-specific attributes such as noise level, Wi-Fi quality, outlet availability, and hours of operation. Users of the system can leave reviews, upload photos, browse amenities, and search for spots by category or location. The database is built using MySQL and follows the relational model, with the design process guided by requirements analysis, ER/EER modeling, relational schema mapping, normalization, and a complete data dictionary.

Key aspects of this design include the many-to-many relationships between study spots and both categories and amenities, the separation of location data into its own relation to eliminate transitive dependencies, and the use of a structured review system that captures both qualitative and quantitative data about each study spot.

2. Requirements Analysis

2.1 Data Requirements

The following entities were identified as necessary to model the study spots application domain. Each entity stores a specific type of information relevant to the system.

User

The User entity stores information about individuals who interact with the system. Users are the primary actors: they write reviews, upload photos, and search for study spots. Every user must be uniquely identifiable.

- UserID (Primary Key) – unique integer identifier for each user
- FirstName – the user's given name
- LastName – the user's family name
- Username – unique display name chosen by the user
- Email – unique email address used for login and contact
- JoinDate – the date the user created their account

StudySpot

The StudySpot entity represents a physical location where people can study. This is the central entity of the database. Each study spot has identifying information, pricing, and references to its location and category.

- SpotID (Primary Key) – unique integer identifier for each study spot
- SpotName – the name of the location
- Description – a brief text description of the study spot
- PhoneNumber – contact phone number for the location
- Website – URL of the study spot's website

- PriceLevel – a numeric rating of price range (1–4)

Category

The Category entity classifies study spots into types. A single study spot may belong to more than one category. For example, a location could be classified as both a cafe and a co-working space.

- CategoryID (Primary Key) – unique integer identifier for each category
- CategoryName – the name of the category (e.g., Cafe, Library, Restaurant)

Location

The Location entity stores physical address and geographic coordinate information for each study spot. Location data is separated from the StudySpot entity to maintain Third Normal Form, since address attributes functionally depend on LocationID rather than SpotID.

- LocationID (Primary Key) – unique integer identifier for each location record
- StreetAddress – street address of the study spot
- City – city where the study spot is located
- State – two-letter state abbreviation
- ZipCode – postal zip code
- Latitude – geographic latitude coordinate
- Longitude – geographic longitude coordinate

Review

The Review entity stores user-submitted evaluations of study spots. Each review is associated with exactly one user and one study spot. Reviews capture both a numeric rating and study-specific environmental attributes.

- ReviewID (Primary Key) – unique identifier for each review
- Rating – integer score from 1 to 10

- ReviewText – free-text written review
- ReviewDate – the date the review was posted
- NoiseLevel – categorical descriptor (e.g., Quiet, Moderate, Loud)
- WiFiQuality – categorical descriptor (e.g., Excellent, Stable, Bad)
- OutletAvailability – boolean indicating whether power outlets are available

Photo

The Photo entity stores images associated with study spots. Users may upload photos to help other users visualize the environment. Each photo is linked to one study spot.

- PhotoID (Primary Key) – unique identifier for each photo
- PhotoURL – the URL path where the photo is stored
- UploadDate – the date the photo was uploaded
- Caption – short optional description of the photo

Hours

The Hours entity stores the operating schedule for each study spot, broken down by day of the week. This allows users to determine whether a spot is open at a given time.

- HoursID (Primary Key) – unique identifier for each hours record
- DayOfWeek – the name of the day (e.g., Monday, Tuesday)
- OpenTime – the time the location opens
- CloseTime – the time the location closes
- IsClosed – boolean indicating whether the location is closed that day

Amenity

The Amenity entity stores features that may be available at study spots. Amenities are shared across multiple locations, so they are stored in a separate entity and linked through a junction table.

- AmenityID (Primary Key) – unique identifier for each amenity
- AmenityName – the name of the amenity (e.g., Wi-Fi, Outdoor Seating, Parking)

2.2 Functional Requirements

The following functional requirements describe the operations the database must support. All requirements are non-trivial and will be implemented in the final system.

1. Users should be able to search for study spots by category (e.g., cafe, library, or restaurant).
2. Users should be able to search for study spots based on available amenities such as Wi-Fi or power outlets.
3. Users should be able to view reviews and ratings for each study spot.
4. Users should be able to find study spots that are open on a particular day and time.
5. Users should be able to search for study spots within a specific city or zip code.
6. Users should be able to view photos associated with each study spot.
7. Users should be able to identify highly rated study spots.
8. Users should be able to leave reviews and ratings for study spots.
9. Users should be able to view operating hours for study spots.
10. Users should be able to compare study spots based on ratings and study-related features such as noise level.

2.3 Relationship Requirements

- A User writes one or more Reviews. Each Review is written by exactly one User.

- A StudySpot receives zero or more Reviews. Each Review belongs to exactly one StudySpot.
- A StudySpot is associated with exactly one Location record.
- A StudySpot has one or more Hours records defining its daily schedule.
- A StudySpot can belong to one or more Categories (many-to-many).
- A StudySpot can have zero or more Photos associated with it.
- A StudySpot can offer multiple Amenities, and each Amenity can apply to many study spots (many-to-many).

2.4 Constraints

- Each user must have a unique UserID and a unique email address.
- Each review must be associated with exactly one user and one study spot.
- Each study spot must have exactly one location record.
- Ratings must fall within the valid range of 1 to 10.
- Each study spot may have multiple photos.
- Hours records must be organized by day of the week.
- Amenities may be shared across multiple study spots.
- JoinDate and ReviewDate cannot be set to future dates.
- The average rating of a study spot may be derived from the associated reviews.

3. ER/EER Model

The conceptual design for the Study Spots database is represented using an Entity-Relationship (ER) diagram. This diagram was developed by first identifying all entities and their attributes (Phase 2.1), then identifying the relationships between them (Phase 2.2), and finally producing a final consolidated diagram with min-max participation constraints (Phase 2.3).

3.1 Entity and Attribute Identification

Eight entities were identified based on the requirements analysis: User, StudySpot, Category, Location, Review, Photo, Hours, and Amenity. Each entity was assigned a primary key and its attributes were diagrammed individually. The primary keys were selected based on the need for stable, unique identifiers that are independent of the data they describe.

For example, UserID was chosen as the primary key for the User entity rather than Email or Username, even though both are unique, because surrogate integer keys perform better as foreign keys in related tables and are not subject to change if a user updates their contact information.

3.2 Relationship Identification

The relationships between entities were identified and diagrammed, including participation constraints. The many-to-many relationships between StudySpot and Category, and between StudySpot and Amenity, are of particular note. These relationships require junction tables in the relational schema. The one-to-many relationships between StudySpot and both Hours and Photo are straightforward, as is the one-to-one relationship between StudySpot and Location.

3.3 Final EER Diagram

The final EER diagram below consolidates all entities and relationships with min-max notation. At minimum, every StudySpot must have exactly one location (1,1) and can have many hours and reviews. Users may have zero reviews. The diagram includes eight entity types and seven relationships, exceeding the minimum requirements of five entity types and four relationships.

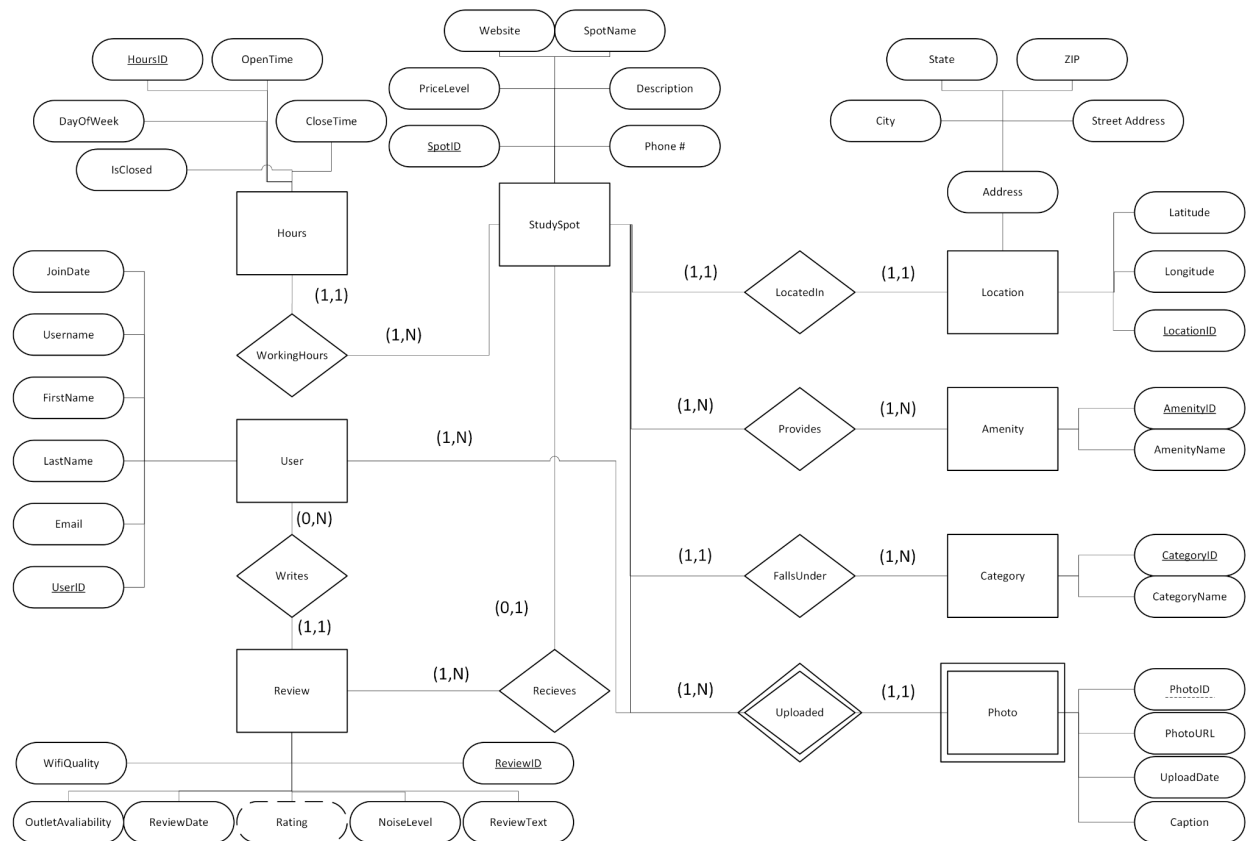


Figure 1. Final EER diagram for the Study Spots database showing all entities, relationships, and participation constraints.

4. Relational Model

In this phase, the EER diagram was mapped to a relational schema following the standard step-by-step mapping procedure. Regular entities were mapped first, followed by weak entities, and then relationships. The most notable mapping decisions involved the many-to-many relationships and the handling of location data.

4.1 Mapping Process

Each regular entity (User, StudySpot, Category, Location, Review, Photo, Hours, Amenity) was mapped directly to a relation, with the primary key of the entity becoming the primary key of the corresponding relation.

The two many-to-many relationships, between StudySpot and Category, and between StudySpot and Amenity, were each mapped to junction tables. StudySpotCategory contains a composite primary key of (SpotID, CategoryID), and StudySpotAmenity contains a composite primary key of (SpotID, AmenityID). These junction tables contain only keys and no additional non-key attributes, which means they are trivially in Third Normal Form.

The one-to-one relationship between StudySpot and Location was handled by placing LocationID as a foreign key in the StudySpot relation. The original design placed all location attributes directly inside StudySpot, but this introduced a transitive dependency that was resolved during the normalization phase by separating location data into its own relation.

The one-to-many relationships, such as StudySpot to Review, StudySpot to Photo, and StudySpot to Hours, were handled by placing SpotID as a foreign key in the child relation (Review, Photo, and Hours respectively). Similarly, the one-to-many relationship from User to Review was handled by placing UserID as a foreign key in the Review relation.

4.2 Final Relational Schema

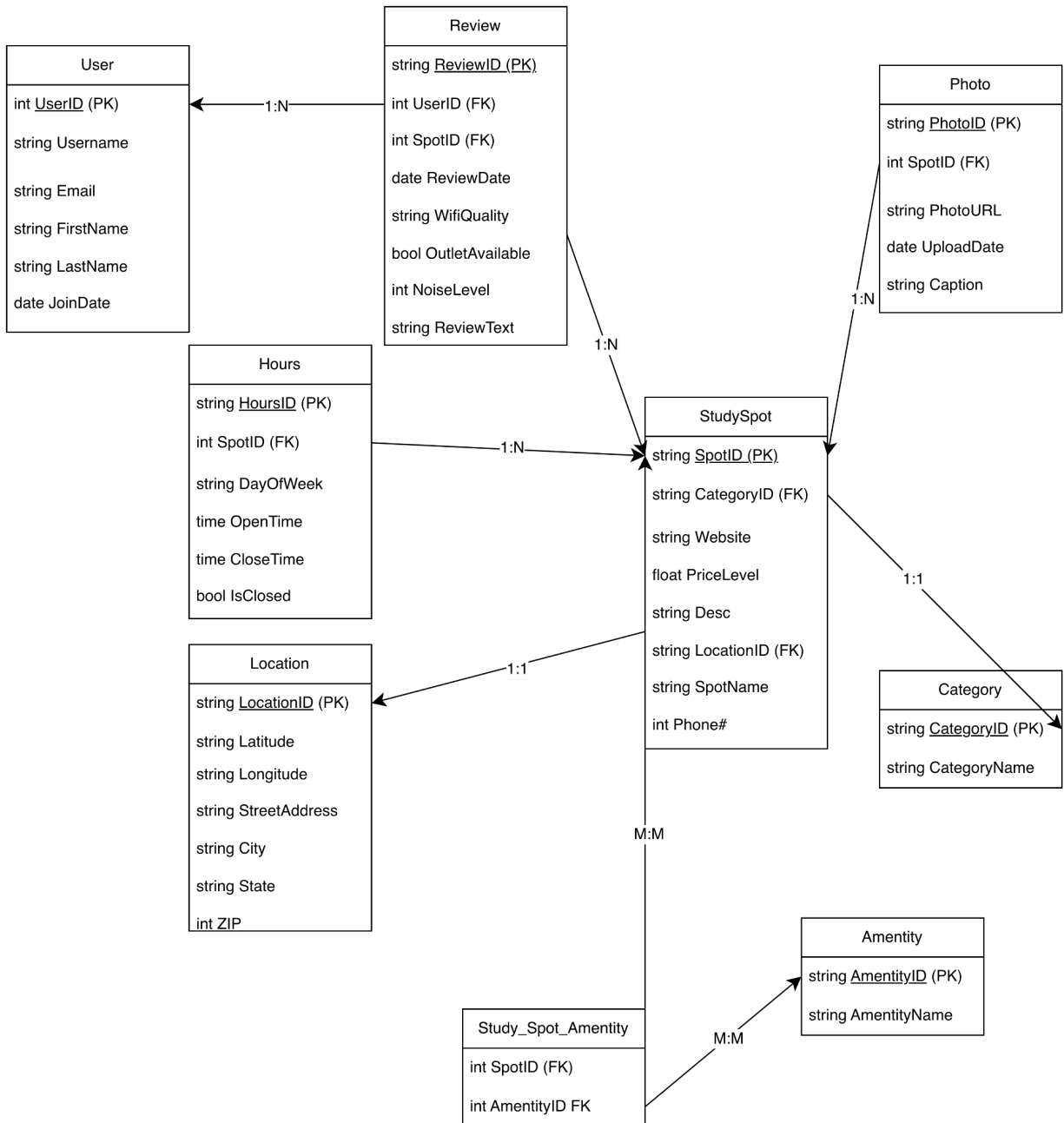


Figure 2. Relational schema diagram showing all relations and their foreign key relationships.

The resulting relational schema is as follows:

User(UserID, FirstName, LastName, Username, Email, JoinDate)

StudySpot(SpotID, LocationID [FK], SpotName, Description, PhoneNumber, Website, PriceLevel)

Location(LocationID, StreetAddress, City, State, Zip, Latitude, Longitude)

Category(CategoryID, CategoryName)

StudySpotCategory(SpotID [FK], CategoryID [FK])

**Review(ReviewID, UserID [FK], SpotID [FK], Rating, ReviewText, ReviewDate,
NoiseLevel, WiFiQuality, OutletAvailable)**

Photo(PhotoID, SpotID [FK], PhotoURL, UploadDate, Caption)

Hours(HoursID, SpotID [FK], DayOfWeek, OpenTime, CloseTime, IsClosed)

Amenity(AmenityID, AmenityName)

StudySpotAmenity(SpotID [FK], AmenityID [FK])

5. Normalization

The objective of this phase is to verify that every relation in the relational schema satisfies the requirements of Third Normal Form (3NF). A relation is in 3NF if it satisfies the following three conditions:

11. It contains only atomic attributes (First Normal Form / 1NF).
12. All non-key attributes are fully functionally dependent on the entire primary key (Second Normal Form / 2NF).
13. No transitive dependencies exist between non-key attributes (Third Normal Form / 3NF).

The Study Spots database was analyzed for functional dependencies throughout each relation. One transitive dependency was identified and resolved through decomposition. The following sections justify the normalization of each relation.

5.1 StudySpot and Location Decomposition

The most significant normalization issue arose in the StudySpot relation. In the original design, location attributes (StreetAddress, City, State, Zip, Latitude, Longitude) were stored directly inside the StudySpot relation alongside LocationID. This created a transitive dependency: $\text{SpotID} \rightarrow \text{LocationID}$, and $\text{LocationID} \rightarrow \text{StreetAddress, City, State, Zip, Latitude, Longitude}$. As a result, the address attributes depended on SpotID only transitively, through LocationID, which violates 3NF.

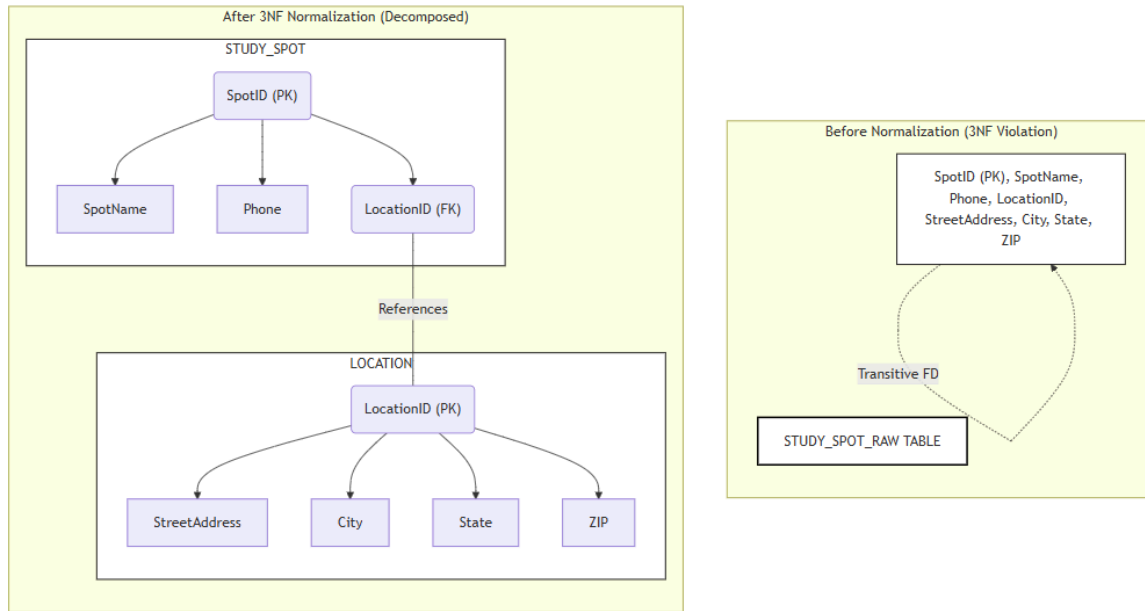


Figure 3. Transitive dependency identified in the original StudySpot relation and its resolution through 3NF decomposition.

To resolve this, the StudySpot relation was decomposed into two separate relations: StudySpot and Location. After decomposition, all non-key attributes in StudySpot depend directly on SpotID, and all non-key attributes in Location depend directly on LocationID. Both relations now satisfy 3NF.

5.2 Category and StudySpotCategory

A study spot may belong to multiple categories, and a single category may apply to many study spots. To prevent repeated CategoryName values across study spots, category information was stored in a separate Category relation. The functional dependency is: CategoryID → CategoryName. The Category relation contains no transitive dependencies and satisfies 3NF. The junction table StudySpotCategory contains only a composite primary key of (SpotID, CategoryID) and no non-key attributes, which trivially satisfies 3NF.

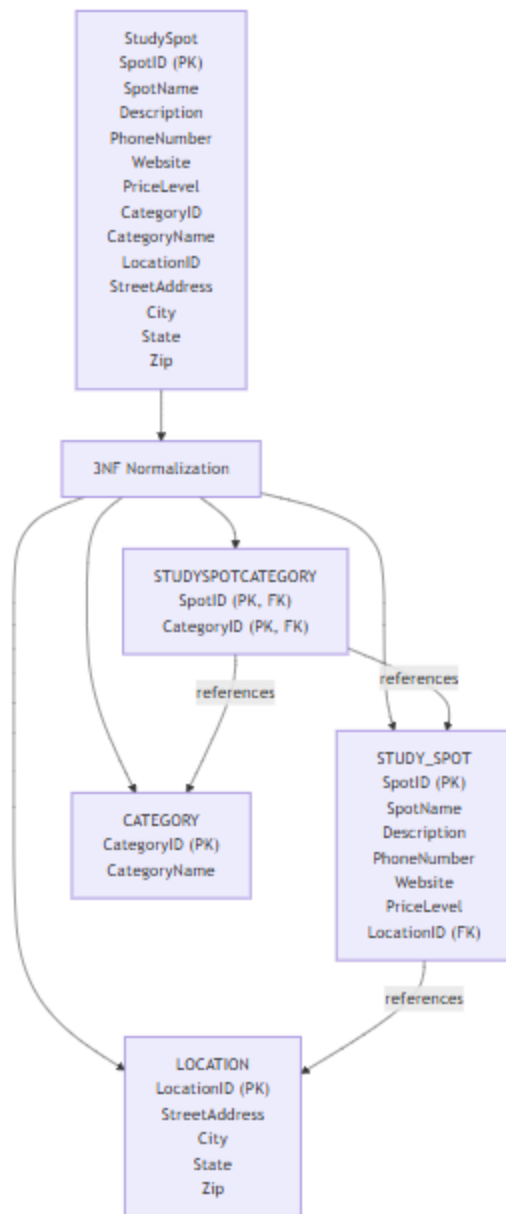


Figure 4. Final 3NF decomposition showing the relationships between StudySpot, Location, Category, and StudySpotCategory.

5.3 Remaining Relations

The remaining relations were each verified to be in 3NF based on the following functional dependencies:

- Review: ReviewID $\rightarrow$ UserID, SpotID, Rating, ReviewText, ReviewDate, NoiseLevel, WiFiQuality, OutletAvailable. All non-key attributes depend directly on ReviewID. No partial or transitive dependencies exist. Review satisfies 3NF.
- User: UserID $\rightarrow$ FirstName, LastName, Username, Email, JoinDate. All non-key attributes depend only on the primary key. User satisfies 3NF.
- Hours: HoursID $\rightarrow$ SpotID, DayOfWeek, OpenTime, CloseTime, IsClosed. All non-key attributes depend only on HoursID. Hours satisfies 3NF.
- Photo: PhotoID $\rightarrow$ SpotID, PhotoURL, UploadDate, Caption. All attributes depend directly on PhotoID. Photo satisfies 3NF.
- Amenity: AmenityID $\rightarrow$ AmenityName. The single non-key attribute depends directly on the primary key. Amenity satisfies 3NF.
- StudySpotAmenity: Contains only a composite primary key of (SpotID, AmenityID) with no non-key attributes. Trivially satisfies 3NF.

In summary, the Study Spots relational schema was analyzed for functional dependencies and decomposed where necessary to eliminate transitive dependencies. The resulting schema satisfies Third Normal Form across all relations, which reduces redundancy, improves consistency, and preserves data integrity.

6. Data Dictionary

The data dictionary below describes the structure and constraints of each table in the Study Spots database. For each attribute, the data type, description, and applicable constraints are specified.

6.1 User

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
UserID	INT	Primary Key; Uniquely identifies each user	Not null; Unique
FirstName	VARCHAR(50)	First name of user	Not null
LastName	VARCHAR(50)	Last name of user	Not null
Username	VARCHAR(50)	Display name of user	Not null; Unique
Email	VARCHAR(50)	Email address of user	Not null; Unique; Follows valid email format

6.2 StudySpot

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
SpotID	INT	Primary key; Uniquely identifies each study spot	Not null; Unique
CategoryID	INT	Foreign key; Uniquely identifies each category; References Category(CategoryID)	Not null
LocationID	INT	Foreign key;	Not null; Unique

		References Location(LocationID)	
SpotName	VARCHAR(100)	Name of spot	Not null
Description	STRING	Description of spot	Null allowed
Phone #	VARCHAR(15)	Phone number of spot	Null allowed
Website	VARCHAR(255)	Website URL of spot	Null allowed
PriceLevel	FLOAT	Price range of spot	Range 1-4

6.3 Category

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
CategoryID	INT	Primary key; Uniquely identifies category	Not null; Unique
CategoryName	VARCHAR(50)	Category type (cafe, library, etc.)	Not null

6.4 Location

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
LocationID	INT	Primary key; Uniquely identifies location	Not null; Unique
StreetAddress	VARCHAR(255)	Street address of spot	Not null
City	VARCHAR(50)	City of spot	Not null
State	VARCHAR(2)	State of spot (ex. CA)	Not null
Zip	INT	ZIP code of spot	Not null
Latitude	FLOAT	Latitude of spot	Null allowed; Valid range -90 to 90

Longitude	FLOAT	Longitude of spot	Null allowed; Valid range -180 to 180
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6.5 Review

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
ReviewID	INT	Primary key; Uniquely identifies a review	Not null; Unique
UserID	INT	Foreign key; Reference from User(UserID)	Not null
SpotID	INT	Foreign key; Reference from StudySpot(SpotID)	Not null
Rating	INT	Rating score given by user	Not null; Range 1-10
ReviewText	STRING	Review written by user	Null allowed
ReviewDate	DATE	Date the review was posted	Not null; Can't be a future date
NoiseLevel	VARCHAR(20)	Noise level of environment (ex. Quiet, Moderate, Loud)	Null allowed

6.6 Photo

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
PhotoID	INT	Primary key; Uniquely identifies photo	Not null; Unique
SpotID	INT	Foreign key;	Not null

		References StudySpot(SpotID)	
PhotoURL	VARCHAR(225)	URL of stored photo	Not null
UploadDate	DATE	Date photo was uploaded	Not null
Caption	VARCHAR(225)	Short description of photo	Null allowed

6.7 Hours

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
HoursID	INT	Primary key; Uniquely identifies time slots	Not null; Unique
SpotID	INT	Foreign Key; References StudySpot(SpotID)	Not null
DayofWeek	VARCHAR(10)	Day name	Not null
OpenTime	TIME	Opening time of spot	Null allowed
CloseTime	TIME	Closing time of spot	Null allowed
IsClosed	BOOLEAN	Indicates if spot is closed	Not null

6.8 Amenity

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
AmenityID	INT	Primary key; Uniquely identifies amenity	Not null; Unique
AmenityName	VARCHAR(50)	Type of amenity (ex. Outlets, outdoor	Not null

		seating)	
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6.9 StudySpotAmenity

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
SpotID	INT	Primary key; Foreign Key; References StudySpot(SpotID)	Not null
AmenityID	INT	Primary key; Foreign key; References Amenity(AmenityID)	Not null

6.10 StudySpotCategory

<u>Attribute</u>	<u>Data Type</u>	<u>Description</u>	<u>Constraints</u>
SpotID	INT	Primary key; Foreign key; References sturdy spot	Not null
CategoryID	INT	Primary key; Foreign key; References category	Not null

7. Summary

This report has documented the full design of the Study Spots database, a relational database system built in MySQL to help users discover, evaluate, and compare study-friendly locations. The design process followed the standard phases of database development: requirements analysis, conceptual modeling, relational schema mapping, normalization, and data dictionary creation.

The database consists of ten relations derived from eight core entity types identified during the requirements and conceptual design phases. The two many-to-many relationships, between study spots and categories and between study spots and amenities, were resolved through the addition of junction tables. One transitive dependency was identified and eliminated during normalization: location address data was separated from the StudySpot relation into its own Location relation to achieve Third Normal Form. All remaining relations were verified to satisfy 3NF based on their functional dependencies.

A key strength of this design is its flexibility. The use of separate Amenity and Category entities with junction tables means new amenities and categories can be added without altering the core schema. The review system captures both numeric ratings and study-specific qualitative attributes like noise level and Wi-Fi quality, making it well-suited to the specific needs of student users. The Hours relation allows fine-grained scheduling information to be recorded by day, enabling users to filter spots by availability.

Potential applications of this database extend beyond a simple search tool. The data collected could be analyzed to identify the most study-friendly neighborhoods in a city, track trends in user preferences over time, or power a recommendation system that matches users to study spots based on their past reviews. The structured schema also makes it straightforward to extend the database in the future, for example by adding a user favorites or bookmarking feature, or by tracking check-ins at study spots.

Overall, the Study Spots database provides a solid foundation for a practical, student-focused application that addresses a real and common need. The design is well-normalized, clearly documented, and ready for implementation.

8. Implementation

The implementation phase translates the normalized relational schema into a fully functional MySQL database. All SQL files are maintained in the project's GitHub repository. The three required SQL files are described below.

GitHub Repo: <https://github.com/chrisalpuerto/cpsc-332-group-project>

8.1 create.sql

This file contains all DDL (Data Definition Language) commands used to create the database schema. It includes CREATE TABLE statements for all ten relations, with primary key and foreign key constraints explicitly defined to enforce referential integrity and data consistency.

<https://github.com/chrisalpuerto/cpsc-332-group-project/blob/main/create.sql>

8.2 insert.sql

This file populates the database with sample data using SQL INSERT statements. The test data was designed to be sufficient to verify all implemented queries, including edge cases such as study spots with no reviews, users with multiple reviews, and spots that belong to multiple categories.

<https://github.com/chrisalpuerto/cpsc-332-group-project/blob/main/insert.sql>

8.3 queries.sql

This file contains all SQL queries corresponding to the use cases identified in the requirements analysis. Each query is properly commented to indicate which functional requirement it addresses. The two primary use cases implemented are:

- Use Case 1: Retrieve all study spots in a given city that have Wi-Fi available and are open on a specified day, along with their average rating, ordered from highest to lowest rated.
- Use Case 2: Retrieve the top-rated study spots by category, showing the spot name, category, average rating, average noise level preference, and total number of reviews for each spot.

<https://github.com/chrisalpuerto/cpsc-332-group-project/blob/main/queries.sql>